AI in Healthcare: The Role of AI in Enhancing Diagnostic Accuracy

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Abstract

Artificial Intelligence (AI) is transforming healthcare by aiding doctors to enhance the accuracy of diagnostics. Studies show that AI outperforms conventional human methods in detecting various diseases. (Saha, Park, Geem, Singh, 2024). AI cannot be used independently; therefore, human-AI collaboration is essential to ensure fair integration into clinical practice.

Problem

- **Diagnostic Mistakes**: Clinicians may occasionally misdiagnose patients due to cognitive bias like heuristics (Li, Cheng, Liu, 2020).
- **Global Inequities**: A lack of specialists in under-resourced areas, shown in Figure 1, causes delays in diagnosing conditions like schizophrenia.
- **Heavy Workload**: Physicians devote significant time to administrative tasks, reducing time for patient care (James, 2023).

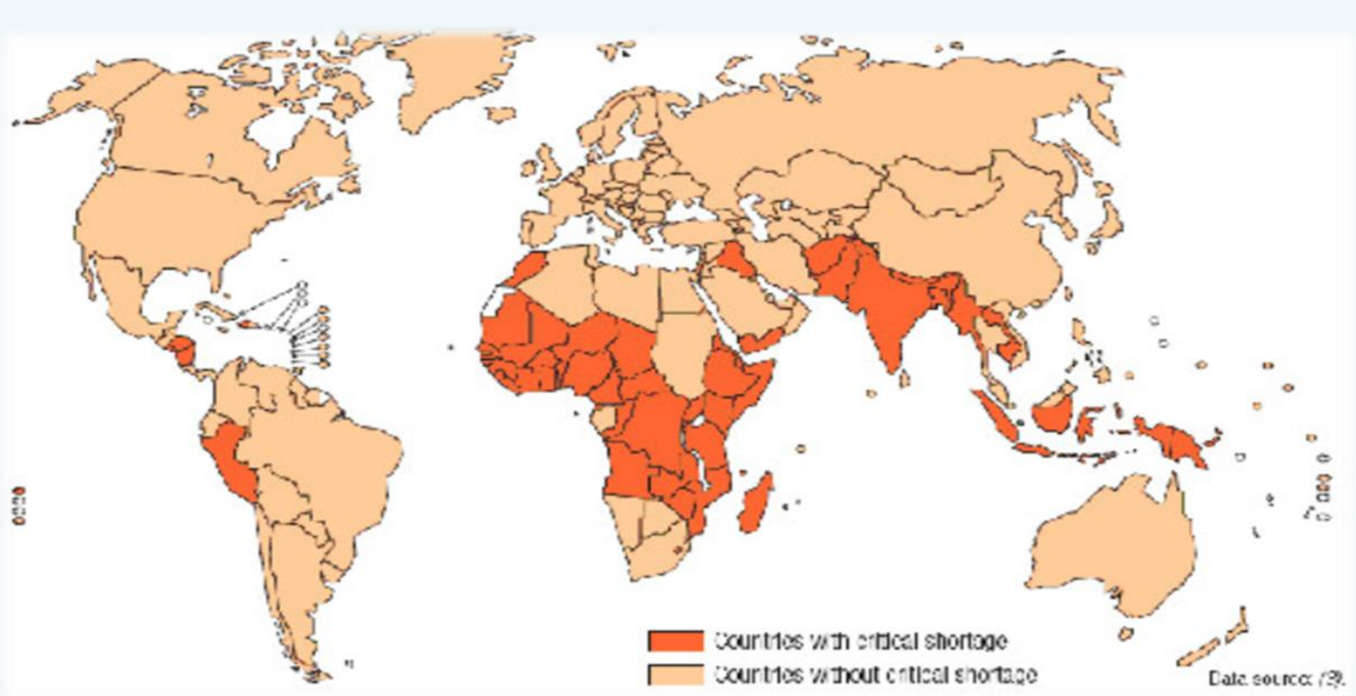

Figure 1. Countries in dark orange have a critical shortage of healthcare providers (World Health Organization, 2006)

Key Applications

- **Schizophrenia Detection**: AI improves schizophrenia detection in EEG analysis, facilitating doctors in detecting the disorder. Figure 2 illustrates how AI accomplishes this.
- **Cancer Detection:** AI-assisted imaging tools enhance radiologists' ability to accurately identify tumors in CT scans (Killock, 2020).

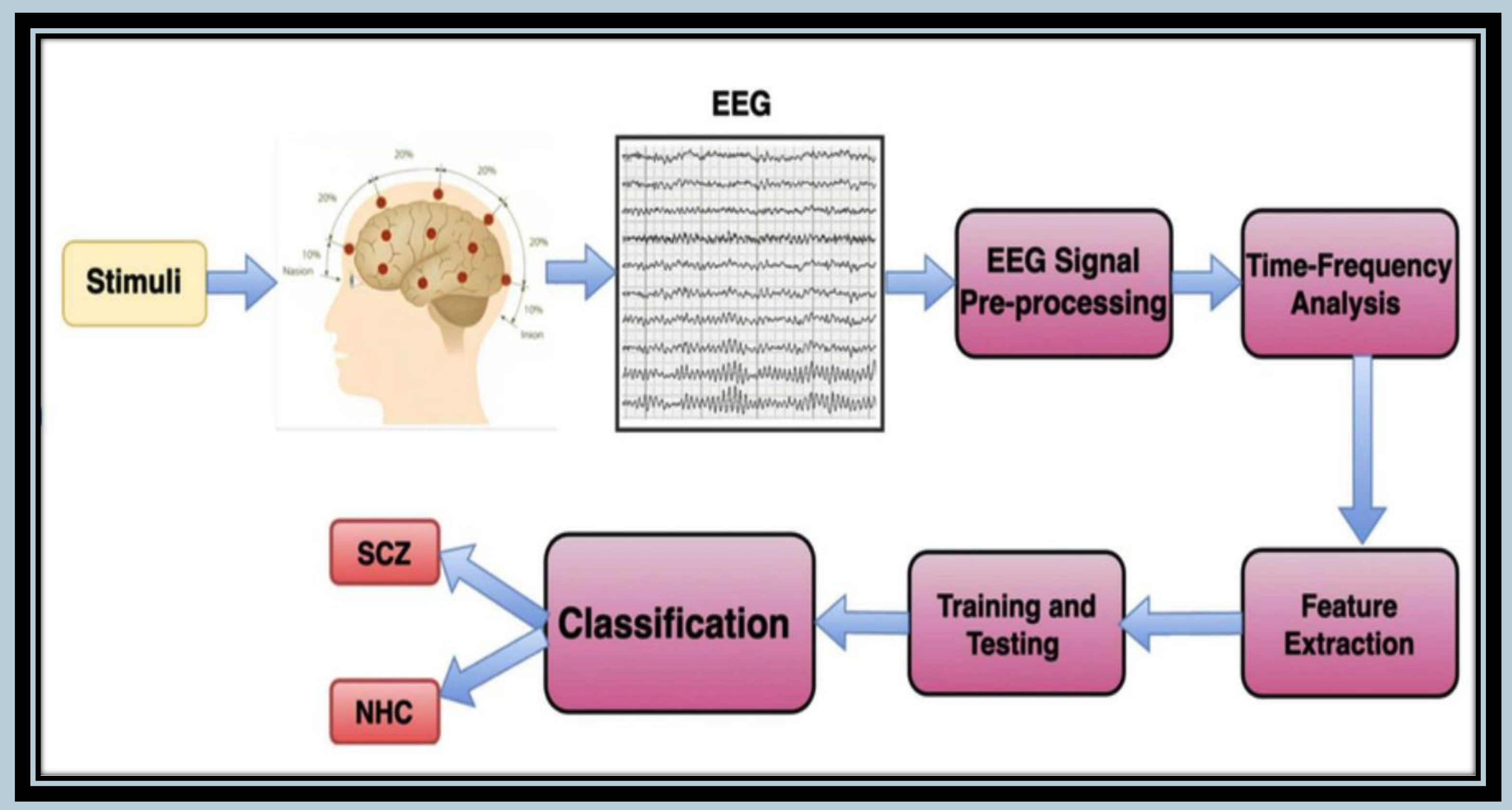

Figure 2. Flow diagram that shows the steps involved in schizophrenia classification using AI (Rahul, Sharma, Sharma, Sarkar, 3).

Benefits

Enhanced Training: Medical students using AI tools report higher diagnostic accuracy (James, 2023).
Continuous Learning: AI can constantly learn and identify new patterns from real-world data to improve its reliability (Brown, 2021).

Challenges

- **Privacy:** AI systems require access to sensitive patient data, which could be vulnerable to data breaches. This is a concern among US adults, as shown in Table 1.
- **Ethical Risks**: Bias in training data can lead to incorrect decisions by the AI, requiring physicians to spend additional time correcting them (Alowais et al., 2023).

	Worse	Better	No Difference
Number of mistakes made by healthcare providers	27%	40%	31%
The quality of healthcare	30%	31%	38%
The security of patients' personal health records	37%	22%	39%
Patients' relationship with their healthcare providers	57%	13%	29%

Table 1. A table summarizing the perspectives of US adults on the potential benefits of AI in healthcare (AIPRM, 2024).

Conclusion

- AI enhances the capabilities of clinicians rather than replacing them, leading to more accurate diagnoses when used in collaboration.
- It is essential to maintain data transparency and establish ethical frameworks to prevent the misuse of patient data.

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